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Self-aligned scheme for semiconductor device and method of forming the same

Abstract

Semiconductor device and the manufacturing method thereof are disclosed herein. An exemplary semiconductor device comprises an interlayer dielectric (ILD) layer disposed over a substrate; a first conductive feature at least partially embedded in the ILD layer; a dielectric layer disposed over and aligned with the ILD layer, wherein a top surface of the dielectric layer is above a top surface of the first conductive feature; an etch stop layer (ESL) disposed over the dielectric layer and over the first conductive feature; and a second conductive feature disposed on the first conductive feature, wherein the second conductive feature includes a first portion having a first bottom surface contacting a top surface of the first conductive feature and a second portion having a second bottom surface contacting a top surface of the dielectric layer.

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Background/Summary

PRIORITY DATA (1) The present application is a continuation application of U.S. patent application Ser. No. 17/749,303, filed May 20, 2022, which is a continuation application of U.S. patent application Ser. No. 16/932,208, filed Jul. 17, 2020, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/906,136, filed Sep. 26, 2019, each of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) The electronics industry has experienced an ever-increasing demand for smaller and faster electronic devices which are simultaneously able to support a greater number of increasingly complex and sophisticated functions. These goals have been achieved by scaling down semiconductor IC dimensions (e.g., minimum feature size) and thereby improving production efficiency and lowering associated costs.

(2) However, aggressive scaling down of IC dimensions has introduced increased complexity to the semiconductor manufacturing process and resulted in some issues of the semiconductor device. For example, aggressive scaling down of IC dimensions, i.e. smaller pitches and critical dimensions, has resulted in difficulties to control the distance between the conductive features, particularly the distance between the different levels of conductive features. For example, it has been observed that due to space dimension shrinkage, a via-to-line breakdown (tiger-tooth) may happen in the back-end of line (BEOL) during IC fabrication. The tiger-tooth issue may affect the device reliability, for example, reduce the time dependent dielectric breakdown (TDDB) and/or increase the parasitic capacitance of the semiconductor device. Thus, improvements in these areas are desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a flowchart of an example method for making a semiconductor device in accordance with some embodiments of the present disclosure;

(3) FIGS. 2-8 illustrate cross-sectional views of the example semiconductor device at intermediate stages of the method of FIG. 1 in accordance with some embodiments of the present disclosure;

(4) FIGS. 9-16 illustrate cross-sectional views of the example semiconductor device at intermediate stages of the method of FIG. 1 in accordance with some other embodiments of the present disclosure; and

(5) FIGS. 17A and 17B illustrate cross-sectional views of block A of FIG. 8 and block B of FIG. 16, respectively, in accordance with some further embodiments of the present disclosure.

DETAILED DESCRIPTION

(6) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature

in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

(7) In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Moreover, the formation of a feature on, connected to, and/or coupled to another feature in the present disclosure that follows may include embodiments in which the features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the features, such that the features may not be in direct contact. In addition, spatially relative terms, for example, “lower,” “upper,” “horizontal,” “vertical,” “above,” “over,” “below,” “beneath,” “up,” “down,” “top,” “bottom,” etc. as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) are used for ease of the present disclosure of one features relationship to another feature. The spatially relative terms are intended to cover different orientations of the device including the features. Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within $\pm 10\%$ of the number described or other values as understood by person skilled in the art. For example, the term “about 5 nm” encompasses the dimension range from 4.5 nm to 5.5 nm.

(8) The present disclosure is generally related to semiconductor devices and the fabrication thereof, and more particularly to forming conductive features (such as metal lines and vias) with self-aligned scheme during the fabrication of an integrated circuit (IC).

(9) Due to the aggressive scaling down of IC dimensions, overlay (mask shifting) issues have been observed in the fabrication of semiconductor devices. During the formation of interconnect layer(s), it is very difficult to control the distance between different levels of conductive features, such as distance between metal line and via, due to the smaller critical dimensions (CD) and the selectivity challenge between the adjacent materials. Via-to-line breakdown may happen between the adjacent conductive features, which may cause manufacturing defects and/or current leakage.

(10) The present disclosure introduces a self-aligned scheme (SAS) for conductive feature formation, such that the via-to-line breakdown issue can be mitigated, and higher pattern density can be achieved. In the present disclosure, a dielectric SAS layer including metal oxide or metal nitride material is deposited over the lower level dielectric layer (for example, an interlayer dielectric (ILD) layer) without touching the lower level conductive feature (for example, a metal line). Thereafter, when forming the higher level conductive feature (for example, a via), the dielectric SAS layer will block the higher level conductive feature from breaking down to the lower level dielectric layer. Accordingly, the higher level conductive feature may include a first portion landing on the lower level conductive feature and a second portion landing on the dielectric SAS layer. Thereby, the distance between the adjacent lower level conductive feature and the higher conductive feature can be increased. In some embodiments, an extra recessing process is performed to the lower level conductive feature to further enlarge the distance between the adjacent lower level conductive feature and the higher level conductive feature. Therefore, the via-to-line breakdown issue may be mitigated, the parasitic capacitance may be reduced, and the performance of the semiconductor device may be improved. Of course, these advantages are merely example, and no particular advantage is required for any particular embodiment.

(11) FIG. 1 illustrates a flow chart of a method **100** for forming a semiconductor device **200** (hereafter called device **200**) in accordance with some embodiments of the present disclosure. Method **100** is merely an example and is not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be performed before, during, and after method **100**, and some operations described can be replaced, eliminated, or moved around for

additional embodiments of the method. Method **100** is described below in conjunction with other figures, which illustrate various cross-sectional views of device **200** during intermediate steps of method **100**. In particular, FIGS. **2-8** illustrate cross-sectional views of the device **200** according to some embodiments of the present disclosure; and FIGS. **9-16** illustrate cross-sectional views of the device **200** during fabrication steps according to some other embodiments of the present disclosure. (12) Device **200** may be an intermediate device fabricated during processing of an integrated circuit (IC), or a portion thereof, that may include static random-access memory (SRAM) and/or other logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as p-type FETs (PFETs), n-type FETs (NFETs), FinFETs, GAA transistors, metal-oxide semiconductor field effect transistors (MOSFET), complementary metal-oxide semiconductor (CMOS) transistors, bipolar transistors, high voltage transistors, high frequency transistors, and/or other memory cells. Device **200** can be a portion of a core region (often referred to as a logic region), a memory region (such as a static random access memory (SRAM) region), an analog region, a peripheral region (often referred to as an input/output (I/O) region), a dummy region, other suitable region, or combinations thereof, of an IC. In some embodiments, device **200** may be a portion of an IC chip, a system on chip (SoC), or portion thereof. The present disclosure is not limited to any particular number of devices or device regions, or to any particular device configurations. For example, though device **200** as illustrated is a three-dimensional FET device (e.g., a FinFET), the present disclosure may also provide embodiments for fabricating planar FET devices. FIGS. **2-16**, **17A** and **17B** have been simplified for the sake of clarity to better understand the inventive concepts of the present disclosure. Additional features can be added in device **200**, and some of the features described below can be replaced, modified, or eliminated in other embodiments of device **200**.

(13) Referring to FIGS. **1** and **2**, at operation **102**, method **100** provides a starting device **200**. In the depicted embodiment of FIG. **2**, device **200** includes a substrate **202**. In the depicted embodiment, substrate **202** is a bulk substrate that includes silicon. Alternatively or additionally, the bulk substrate includes another elementary semiconductor, such as germanium; a compound semiconductor, such as silicon carbide, silicon phosphide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, indium antimonide, zinc oxide, zinc selenide, zinc sulfide, zinc telluride, cadmium selenide, cadmium sulfide, and/or cadmium telluride; an alloy semiconductor, such as SiGe, SiPC, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; other group III-V materials; other group II-IV materials; or combinations thereof. Alternatively, substrate **202** is a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GOI) substrate. Semiconductor-on-insulator substrates can be fabricated using separation by implantation of oxygen (SIMOX), wafer bonding, and/or other suitable methods. Substrate **202** may include various doped regions. In some examples, substrate **202** includes n-type doped regions (for example, n-type wells) doped with n-type dopants, such as phosphorus (for example, .sup.31P), arsenic, other n-type dopant, or combinations thereof. In the depicted implementation, substrate **202** includes p-type doped region (for example, p-type wells) doped with p-type dopants, such as boron (for example, .sup.11B, BF.sub.2), indium, other p-type dopant, or combinations thereof. In some embodiments, substrate **202** includes doped regions formed with a combination of p-type dopants and n-type dopants. The various doped regions can be formed directly on and/or in substrate **202**, for example, providing a p-well structure, an n-well structure, a dual-well structure, a raised structure, or combinations thereof. An ion implantation process, a diffusion process, and/or other suitable doping process can be performed to form the various doped regions.

(14) Device **200** also includes a structure **204** disposed over substrate **202**. Structure **204** is a part of a multilayer interconnect feature (MIF) configured to connect the various features of device **200** to form a functional circuit. In some embodiments, structure **204** may be a part of an interlayer dielectric (ILD) layer which may include a dielectric material, such as SiO₂, SiON, TEOS formed

oxide, PSG, BPSG, a low-k dielectric material ($K < 3.9$), or combinations thereof. In some other embodiments, structure **204** may be a conductive structure which is part of an electrode of a transistor, such as a source electrode, a drain electrode, or a gate electrode. A source (or drain) electrode may include n-type doped silicon for NFETs, p-type doped silicon germanium for PFETs, or other suitable materials. A source (or drain) electrode may also include silicide such as nickel silicide, titanium silicide, cobalt silicide, or other suitable silicidation on or germanosilicidation. A gate electrode may include aluminum, tungsten, cobalt, and/or other suitable materials. In some other embodiments, structure **204** may be a conductive structure which is part of a contact feature such as a source contact, a drain contact, or a gate contact, and may include cobalt, tungsten, ruthenium, rhodium, iridium, molybdenum, other metals, a metal nitride such as titanium nitride or tantalum nitride, or a combination thereof. In some embodiments, structure **204** may be a conductive structure which is part of an interconnect structure, such as a metal wire or a metal plug, and may include copper, cobalt, tungsten, ruthenium, rhodium, iridium, molybdenum, other metals, a metal nitride such as titanium nitride or tantalum nitride, or a combination thereof.

(15) Device **200** also includes a first etch stop layer (ESL) **206** disposed over structure **204**. In some embodiments, the first ESL **206** includes a dielectric material, such as a material that includes silicon, oxygen, and/or nitrogen. For example, the first ESL **206** may include aluminum oxide (AlOx), aluminum oxy-nitride (AlON), silicon carbide (SiC), silicon dioxide (SiO₂), silicon oxy-carbide (SiOC), silicon nitride (SiN), silicon carbonitride (SiCN), silicon oxy-nitride (SiON), silicon oxy-carbonitride (SiOCN), other dielectric material, or combinations thereof. The first ESL **206** may be formed by a deposition process, such as chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), other suitable methods, or combinations thereof. The first ESL **206** is optional and may be omitted in some embodiments.

(16) Device **200** also includes a first interlayer dielectric (ILD) layer **208** formed over substrate **202**. The first ILD layer **208** includes a dielectric material such as SiO, SiON, TEOS formed oxide, PSG, BPSG, low-k dielectric material ($K < 3.9$), other suitable dielectric material, or combinations thereof. Exemplary low-k dielectric materials include FSG, carbon doped silicon oxide, Black Diamond® (Applied Materials of Santa Clara, California), Xerogel, Aerogel, Parylene, BCB, SILK (Dow Chemical, Midland, Michigan), polyimide, other low-k dielectric material, or combinations thereof. The first ILD layer **208** may include a multi-layer structure having multiple dielectric materials and may be formed by a deposition process such as CVD, (for example, plasma enhanced CVD (PECVD), flowable CVD (FCVD)), spin-on-glass (SOG), other suitable methods, or combinations thereof. Subsequent to the deposition of the first ILD layer **208**, a CMP process and/or other planarization process is performed to planarize the top surface of device **200**.

(17) Device **200** also includes first conductive features **210A** and **210B** (both referred to as first conductive features **210**). The first conductive features **210** are also referred to as lower level conductive features, which may be conductive interconnect structures such as contacts, vias, or metal lines, such that conductive connections can be set up between the semiconductor components of device **200**. In the depicted embodiment, the first conductive features **210** are metal lines (also referred to as metal lines **210**). In some embodiments, metal lines **210** include a conductive material such as Ta, TaN, TiN, Cu, Co, Ru, Mo, W, other conductive material, or combinations thereof. In some embodiments, metal lines **210** are formed by PVD, CVD, ALD, plating, or other deposition process. In some embodiments, metal lines **210** are formed by a single damascene process or a dual damascene process. For example, in a single damascene process, first, a mask layer with a negative pattern (i.e. a pattern that is opposite to the pattern of metal lines **210**) is deposited over the first ILD layer **208**. The first ILD layer **208** is then patterned (by etching) along the mask layer to form metal line trenches therein. After removing the mask layer, a barrier layer **212**, including a material such as Ta, TaN, Ti, TiN, other suitable material, or combinations thereof, is deposited over the patterned first ILD layer **208** and within the metal line trenches. Barrier layer **212** can provide diffusion barrier properties, which prevent diffusion of the conductive material

(for example Cu) of metal lines **210** into the first ILD layer **208**. Thereafter, a seed layer (not shown), including the conductive material of the metal line, is deposited over barrier layer **212**. Subsequently, the conductive material of via **226** and metal line **228** is formed (for example, by electrochemical plating (ECP)) over the seed layer. And, a planarization process, such as CMP, is performed to device **200** to remove the excessive conductive material and to expose the top surface of the first ILD layer **208**. The remained conductive material forms metal lines **210** (including metal lines **210A** and **210B**). Referring to FIG. 2, metal lines **210** and barrier layer **212** are exposed at the top surface of device **200**. In the depicted embodiments, metal lines **210A** and **210B** extend in parallel in their lengthwise direction, that is along the y-direction.

(18) Referring to FIGS. 1 and 3, at operation **106**, a capping layer **214** is formed over metal lines **210**. As depicted in FIG. 3, capping layer **214** is formed on and aligned with the respective metal lines **210A** and **210B** as well as barrier layer **212**. In other word, capping layer **214** is formed on the conductive materials (metal lines **210** and barrier layer **212**) at the top surface of device **200**, not the dielectric top surface of the first ILD layer **208**. In some embodiment, capping layer **214** includes a conductive material, such as cobalt (Co). The Co capping layer **214** can protect metal lines **210** (for example, including Cu) from being oxidized and can increase the reliability of the Cu metal lines **210**. In some embodiments, capping layer **214** is formed by a selective CVD, other selective metallic capping process, or combinations thereof. In some embodiments, a thickness T1 of capping layer **214** in the z-direction is about 2.5 nm to about 3.5 nm.

(19) Referring to FIGS. 1 and 4, at operation **108**, a blocking layer **216** is formed over capping layer **214**. As depicted in FIG. 4, blocking layer **216** is coated on and aligned with capping layer **214**. In other words, blocking layer **216** does not cover the top surface of the first ILD layer **208**. In some embodiments, blocking layer **216** includes a self-assembling monolayer (SAM) material including hydrophobic head and hydrophobic tail groups. For example, blocking layer **216** comprises chemicals including phosphate group or high-nitrogen (hi-N) group. Blocking layer **216** may be an organic layer, a self-cross-link layer, a self-adhesion layer, other suitable layer, or combinations thereof. In some embodiments, the SAM blocking layer **216** is coated on capping layer **214** by a chemical adsorption process, such that the chemicals of blocking layer **216** is adsorbed only by the metal surface of capping layer **214**, not the dielectric material of the first ILD layer **208** due to the different polarity. As depicted in FIG. 4, blocking layer **216** has a thickness T2 in the z-direction. Thickness T2 can be modulated depending on the distance D1 or D2 (see FIGS. 8, 9, 17, and 18) to be enlarged between the later formed higher level conductive feature (for example, via **226**) and the adjacent lower level conductive feature (for example, metal line **210B**). In some embodiments, thickness T2 is about 1.5 nm to about 4 nm.

(20) Referring to FIGS. 1 and 5, at operation **110**, a dielectric layer **218** is formed over the first ILD layer **208**. Dielectric layer **218** works as a self-align scheme layer (thus is also referred to as SAS layer **218**) during the fabrication of the higher level conductive feature (for example via **226**). SAS layer **218** can stop the tiger-tooth portion of the higher level conductive feature from breaking down to the first ILD layer **208**, thereby to increase the distance between the higher level conductive feature (via **226**) and the adjacent lower level conductive feature (metal line **210B**) and mitigate the current leakage issue therebetween. As depicted in FIG. 5, SAS layer **218** is formed along sidewalls of capping layer **214** and blocking layer **216**. In other words, SAS layer **218** is formed on and aligned with the first ILD layer **208**, since it is blocked by blocking layer **216** formed on and aligned with metal lines **210**. Referring to FIG. 5, SAS layer **218** has sidewall surfaces contacting sidewall surfaces of capping layer **214** and blocking layer **216**. SAS layer **218** includes a material that can provide a different etching rate than that of the later formed second ESL **220** and the second ILD **222** (see, FIGS. 8 and 9), such that during the formation of the via trench, SAS layer **218** can remain substantially unchanged due to the different etching selectivity. In some embodiments, SAS layer **218** includes a metal oxide or metal nitride material, such as aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (Al.sub.2N.sub.3), aluminum oxynitride

(AlON), other suitable dielectric material, or combinations thereof. In some embodiments, SAS layer **218** is formed by a selective ALD process, such that it only forms on the top surface of the first ILD layer **208**. As mentioned above, the first ILD layer **208** comprises low-k dielectric material (for example, SiO.sub.2) and blocking layer **216** comprises organic compound. The material of SAS layer **218** (for example, Al.sub.2O.sub.3, Al.sub.2N.sub.3, and/or AlON) only bond with the low-k material of the first ILD layer **208**, not the SAM organic compound of blocking layer **216**, due to the different chemical affinity. Referring to FIG. 5, SAS layer **218** has a thickness **T3** which is greater than the thickness **T1** of capping layer **214** in the z-direction. Similar as blocking layer **216**, thickness **T3** of SAS layer **218** can be modulated depending on the distance **D1** or **D2** (FIG. 8 or 16) to be enlarged between the later formed higher level conductive feature (for example, vias **226**) and the adjacent lower level conductive feature (for example, metal line **210B**). In some embodiments, thickness **T3** is about 3 nm to about 5 nm.

(21) Referring to FIGS. 1 and 6, at operation **112**, blocking layer **216** is removed, such that capping layer **214** is exposed at the top surface of device **200**. In some embodiments, blocking layer **216** is removed by a hydrogen (H.sub.2) treatment. For example, a reacting gas including H.sub.2 (with a pressure of about 1 torr to about 3 torr) is applied to the top surface of device **200**, the organic material of blocking layer **216** reacts with the H.sub.2 gas, thus blocking layer **216** can be removed. In some other embodiments, blocking layer **216** is removed by a nitrogen (for example, N.sub.2 or NH.sub.3) treatment at a temperature of about 250° C. to about 400° C. As depicted in FIG. 6, after removing blocking layer **216**, SAS layer **218** includes sidewall surfaces contacting and extending above sidewall surfaces of capping layer **214**.

(22) Referring to FIGS. 1 and 7, at operation **114**, a second ESL **220** is deposited over device **200**, in particular, over SAS layer **218** and capping layer **214**. As depicted in FIG. 7, due to the thickness difference between SAS layer **218** and capping layer **214** (thickness **T3** of SAS layer **218** is greater than thickness **T1** of capping layer **214**), the second ESL **220** extends from the top surfaces of SAS layer **218** to the top surfaces of capping layer **214**. In the depicted embodiment, the second ESL **220** includes first portions **220-1** contacting portions of the sidewall surfaces of SAS layer **218** and second portions **220-2** covering the top surfaces of SAS layer **218** and the top surfaces of capping layer **214**. The second ESL **220** includes a dielectric material that has a different etching rate than the dielectric material of SAS **218**. In some embodiments, the second ESL **220** includes a dielectric material such as a material that includes silicon, oxygen, and/or nitrogen. For example, the second ESL **220** may include silicon carbide (SiC), silicon dioxide (SiO.sub.2), silicon oxy-carbide (SiOC), silicon nitride (SiN), silicon carbonitride (SiCN), silicon oxy-nitride (SiON), silicon oxy-carbonitride (SiOCN), other dielectric material, or combinations thereof. The second ESL **220** may be formed by a deposition process, such as atomic layer deposition (ALD), other suitable methods, or combinations thereof. A thickness **T4** of the second ESL **220** in the z-direction may be modulated due to the requirements of resistance/capacitance performance of device **200** and to mitigate the hillock issue thereof. In some embodiments, thickness **T4** of the second ESL **220** in the z-direction is about 1 nm to about 3 nm.

(23) Now referring to FIGS. 1 and 8, at operations **116** and **118**, more conductive features (i.e. higher level conductive feature(s)), for example, a via **226** and a metal line **228** are formed over one of the first conductive features **210** (i.e. a lower level conductive feature). The higher level conductive features can be formed by single damascene process, dual damascene process, other suitable process, or combinations thereof. For example, in a dual damascene process, at operation **116**, a second ILD layer **222** is formed over the second ESL **220**. The material of the second ILD layer **222** is different than that of SAS layer **218** to provide a different etching rate in the later metal line trench **223** and via trench **224** formation process. The fabrication of the second ILD layer **222** is similar as that of the first ILD **208**.

(24) Still referring to FIG. 8, at operation **118**, a metal line trench **223** and a via trench **224** is patterned in the second ILD layer **222**. Metal line trench **223** and via trench **224** form a “T” shape

in the X-Z plane. Metal line trench **223** and via trench **224** can be formed by etching processes (such as dry etching, wet etching, or a combination thereof) via one or more patterned hard masks. The width and depth of the trenches are based on the design requirements of the device **200**. Due to the small IC dimensions and the process restriction, a shifting may happen during the formation of metal line trench **223** and/or via trench **224**. As depicted in FIG. 8, via trench **224** includes a main portion (the left side of the dashed line) landed over the metal line **210A** and a shifting portion (the right side of the dashed line) landed over SAS layer **218**.

(25) Thereafter, a barrier layer **230**, including a material such as Ta, TaN, Ti, TiN, other suitable material, or combinations thereof, is deposited in metal line trench **223** and via trench **224**.

Thereafter, a seed layer (not shown), including the conductive material of the via and/or the metal line, is deposited over barrier layer **230**. Subsequently, the conductive material of via **226** and metal line **228** is formed (for example, by electrochemical plating (ECP)) over the seed layer in metal line trench **223** and via trench **224**. The conductive material may include Ta, Ti, Al, Cu, Co, W, TiN, TaN, other suitable conductive materials, or combinations thereof. And, a planarization process, such as CMP, is performed to device **200** to remove the excessive conductive material and to expose the top surface of the second ILD layer **222**. The conductive material in via trench **224** forms via **226** and the remained conductive material in metal line trench **223** forms metal line **228**. As depicted in FIG. 8, via **226** includes a first portion **226-1** landed over metal line **210A** and a second portion **226-2** landed over SAS layer **218**. In some embodiments, a shifting ration, i.e. a ration of a width $W1$ of the second portion **226-2** to a width $W2$ of via **226** is about 15% to about 30%. Here, the width $W1$ and the width $W2$ are the average width of the second portion **226-2** and the average width of via **226** in the X-direction, respectively.

(26) As discussed above, due to the small critical dimensions of device **200**, it is difficult to ensure that the via trench is perfectly formed over metal line **210A**, and overlay shifting issue often happens during the formation of the via trench. Thus, the via trench may include a main portion and a shifting portion. In a conventional semiconductor structure where no SAS layer **218** is formed, due to the selectivity challenge of the ILD layer and the ESL, materials of the ILD layer and the ESL may be removed together during formation of the via trench. Thus, the shifting portion of via trench may punch through the second ESL and inserting into the first ILD layer. Therefore, the later formed via including two portions, i.e. a first portion over the lower level metal line and a second portion formed besides the first portion and inserting into the first ILD layer. This second portion of the via is also called a tiger-tooth portion, which reduce the distance between the via and the adjacent lower level adjacent metal line. In some critical situations, the tiger-tooth portion of the higher level conductive feature may form breakdown path and induce current leakage to the adjacent lower level conductive feature.

(27) However, in the present disclosure, a SAS layer **218** is formed between the first ILD layer **208** and the second ESL **220**. Due to the different etching selectivity of SAS layer **218** and the second ESL **220**, the shifting portion of the via trench stops at SAS layer **218**. Therefore, as depicted in FIG. 8, via **226** in the present disclosure includes two portions, i.e. a first portion **226-1** over metal line **210A** and a second portion **226-2** landing on SAS layer **218**. In other words, SAS layer **218** stops via **226** from punching into the first ILD layer **208**. Thus, a distance $D1$ between via **226** (the higher level conductive feature) and the adjacent metal line **210B** (the adjacent lower level conductive feature) is ensured. The breakdown path and/or current leakage issues are mitigated and reliability of device **200** is increased. The fabrication of the second conductive feature in the present disclosure is referred to as a self-aligned scheme.

(28) Now referring to FIG. 1, at operations **120**, further processes are performed to complete the fabrication of device **200**. For example, it may form various contacts/vias, wires, and multilayer interconnect features (such as ILD layer(s), ESL(s)) over device **200** according to the design requirement of device **200**.

(29) Method **100** may further include an operation **104** between operation **102** (receiving a starting

semiconductor device with a first conductive feature formed in a first ILD layer) and operation **106** (forming a capping layer on the first conductive feature). FIGS. **9-16** illustrate cross-sectional views of device **200** at intermediate stages of method **100** including operation **104**. Same reference numbers in FIGS. **9-16** refer to the same components/structures of device **200** as those in FIGS. **2-8**, which include same materials and are formed by the same fabrication processes as aforementioned. The critical dimension of the same component/structure is also the same as aforementioned unless it is explicitly recited below.

(30) Referring to FIGS. **1** and **9**, a starting semiconductor device **200** is received. Starting device **200** in FIG. **10** includes the same components/structures as that in FIG. **2**.

(31) Referring to FIGS. **1** and **10**, at operation **104**, top portions of the first conductive features **210** (metal lines **210A** and **210B**) are recessed such that top surfaces of the first conductive features **210A** and **210B** are below the top surface of the first ILD layer **208**. In some embodiments, top portions of the first conductive features **210** are removed by wet cleaning. Due to the different materials of barrier layer **212**, the first conductive features **210**, and the first ILD layer **208**, the wet cleaning process only removes the top portions of the first conductive features **210**, while barrier layer **212** and the first ILD layer **208** remain substantially unchanged. In some embodiments, device **200** is soaked in a wet etchant, such as diluted HF, or other suitable chemicals, or combinations thereof. The extent of the recession may be tunable or quantitatively controllable via the adjustment of various process conditions such as time and temperature. In some embodiments, a recession extent T5 of the first conductive features **210** is about 10% to about 20% of a height T6 of the first conductive features **210**. In some further embodiments, thickness T5 is about 2 nm to about 5 nm.

(32) Referring to FIGS. **1** and **11**, at operation **106**, a capping layer **214** is deposited over the first conductive features **210**. In some embodiments, capping layer **214** may be considered as a portion of the first conductive feature **210**. As depicted in FIGS. **11** and **12**, the top surface of capping layer **214** is below the top surface of the first ILD layer **208**.

(33) Referring to FIGS. **1** and **12**, at operation **108**, a blocking layer **216** is formed over the conductive capping layer **214**, without touching the dielectric material of the first ILD layer **208**. As depicted in FIGS. **12** and **13**, the top surface of blocking layer **216** is above the top surface of the first ILD layer **208**.

(34) Referring to FIGS. **1** and **13**, at operation **110**, a dielectric SAS layer **218** is formed over the first ILD layer **208** and is distanced away from the first conductive features **210**, as well as barrier layer **212**. Similar as discussed above, due to the different chemical affinity, SAS layer **218** does not bond with blocking layer, such that it only formed on and aligned with the first ILD layer **208**.

(35) Referring to FIGS. **1** and **14**, at operation **112**, blocking layer **216** is removed, such that the top surface of capping layer **214** and the top surface and at least portions of the sidewall surfaces of barrier layer **212** are exposed from the top portion of device **200**.

(36) Referring to FIGS. **1** and **15**, at operation **114**, a second ESL **220** is deposited over device **200**. As depicted in FIG. **15**, due to the thickness difference between SAS layer **218** and barrier layer **212** and between barrier layer **212** and capping layer **214**, the second ESL **220** forms steps over SAS layer **218**, barrier layer **212**, and capping layer **214**. The second ESL **220** extends from the top surfaces of SAS layer **218**, along sidewall surfaces of SAS layer **218**, further extends over the top surfaces and along portions of sidewall surfaces of barrier layer **212**, and further extends over the top surfaces of capping layer **214**. Referring to FIG. **15**, the second ESL **220** includes three portions, i.e. first portions **220-1** contacting the sidewall surfaces of SAS layer **218**, second portions **220-2** covering the top surfaces of SAS layer **218** and the top surfaces of capping layer **214**, and third portions **220-3** contacting portions of the sidewall surfaces of barrier layer **212**.

(37) Referring to FIGS. **1** and **16**, at operations **116** and **118**, more conductive features, i.e. via **226'** and metal line **228**, are formed over the first conductive feature **210A** (metal line **210A**). The formation processes and material of via **226'** and metal line **228** are similar as via **226** and metal

line **228** in FIG. **8**. In the example of FIG. **16**, via **226'** includes a first portion **226'-1** disposed metal line **210A** and a second portion (tiger-tooth portion) **226'-2** landing on SAS layer **218**. In some embodiments, a shifting ration, i.e. a ration of a width $W1'$ of the second portion **226'-2** to a width $W2'$ of via **226'** is about 15% to about 30%. Here, the width $W1'$ and the width $W2'$ are the average width of the second portion **226'-2** and the average width of via **226'** in the X-direction, respectively. Due to the recessing process at operation **104**, each of the first conductive features (metal lines) **210A** and **210B** has a top surface below the bottom surface of SAS layer **218** (i.e. the top surface of the first ILD layer **208**). Therefore, the distance $D2$ between the tiger-tooth portion **226'-2** of via **226'** and the adjacent metal line **210B** in FIG. **16** is greater than the distance $D1$ between the tiger-tooth portion **226-2** of via **226** and the adjacent metal line **210B** in FIG. **8**, both are greater than the distance between the tiger-tooth portion of the via and the adjacent lower level metal line in a conventional semiconductor device. Compare the embodiments of FIG. **16** and FIG. **8**, due to the increased distance $D2$, the breakdown path/current leakage issues are further mitigated, the reliability of semiconductor device is further increased, the via-to-line parasitic capacitance is reduced, and the speed of device **200** is boosted.

(38) Referring to FIG. **1**, at operations **120**, further processes may be performed to complete the fabrication of device **200**.

(39) FIGS. **17A** and **17B** illustrate cross-section views of some other embodiments of device **220**. Only portions of device **200** in block A of FIG. **8** and block B of FIG. **16** are illustrated in FIGS. **17A** and **17B**, respectively. As depicted in each of FIGS. **17A** and **17B**, the second ESL **220** includes a multi-layer structure. For example, in FIG. **17A**, the second ESL **220** includes a first ESL film **220-A** having a portion contacting the sidewall surfaces of SAS layer **218** and forming a step extending from the top surface of SAS layer **218** to the top surface of capping layer **214**, and a second ESL film **220-B** disposed over the first ESL film **220-A**. The first ESL film **220-A** may be formed by ALD, and the second film **220-B** may be formed by CVD and/or PVD. The material of the second ESL film **220-B** may be merged over the top of the first conductive feature **210A** during the CVD or PVD process, thus may form a substantially flat top surface. Thereby, the height difference of the first ESL film **220-A** is flattened by the second ESL film **220-B**. In some embodiments, the second ESL layer **220** may also include more ESL films, such as a third ESL film **220-C** as depicted in FIG. **17A**, deposited over the previous ESL film by CVD, PVD, ALD, other suitable process, or combinations thereof. Similarly, in FIG. **17B**, the second ESL **220** may include a first ESL film **220-A** having first portion(s) contacting the sidewall surface(s) of SAS layer **218**, second portion(s) contacting portions of the sidewall surface(s) of barrier layer **212**, and forming steps extending from the top surface of SAS layer **218** to the top surfaces of barrier layer **212**, and further to the top surface of capping layer **214**. The second ESL **220** may also include a second ESL film **220-B** disposed over the first ESL film **220-A**. The first ESL film **220-A** may be formed by ALD, and the second film **220-B** may be formed by CVD and/or PVD, such that the height difference of the first ESL film **220-A** is flattened by the second ESL film **220-B**. In some embodiments, the second ESL layer **220** may also include more ESL films, such as a third ESL film **220-C** as depicted in FIG. **17B**, deposited over the previous ESL film by CVD, PVD, ALD, other suitable process, or combinations thereof.

(40) Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to a semiconductor device and a formation process thereof. For example, embodiments of the present disclosure provide a semiconductor device with conductive interconnect structure(s) formed with a self-aligned scheme. In the present disclosure, a dielectric SAS layer is formed between the upper level ESL (for example, the second ESL **220**) and the lower level ILD layer (for example, the first ILD layer **208**). The SAS layer includes a dielectric material that is different from the material of the upper level ESL and the upper level ILD, such that the SAS layer may remain substantially unchanged during the formation of the upper level conductive feature. Thereby, the SAS layer can stop the upper level conductive feature from breaking down to

the lower level ILD layer, thereby the distance between the upper level conductive feature (for example, via **226**) and the adjacent lower level conductive feature (for example, metal line **210B**) is enlarged. Therefore, the semiconductor device has better reliability due to the reducing of the breakdown path/current leakage issues. In some embodiments, a recessing process is performed to the lower level conductive feature to further enlarge the distance between the upper level conductive feature and the adjacent lower level conductive feature, thereby to reduce the parasitic capacitance and further improve the performance of the semiconductor device. The fabrication process can be integrated in the current process flow and can be applied to multiple technology generations.

(41) The present disclosure provides for many different embodiments. An exemplary semiconductor device comprises an interlayer dielectric (ILD) layer disposed over a substrate; a first conductive feature at least partially embedded in the ILD layer; a dielectric layer disposed over and aligned with the ILD layer, wherein a top surface of the dielectric layer is above a top surface of the first conductive feature; an etch stop layer (ESL) disposed over the dielectric layer and over the first conductive feature; and a second conductive feature disposed on the first conductive feature, wherein the second conductive feature includes a first portion having a first bottom surface contacting a top surface of the first conductive feature and a second portion having a second bottom surface contacting a top surface of the dielectric layer.

(42) In some embodiments, the semiconductor device further comprises a capping layer disposed on and aligned with the first conductive feature. In some embodiments, a thickness of the dielectric layer is greater than a thickness of the capping layer such that the dielectric layer includes a sidewall surface extended above the capping layer. In some embodiments, the ESL includes a portion contacting the sidewall surface of the dielectric layer and wherein the ESL extends from the top surface of the dielectric layer to a top surface of the capping layer. In some embodiments, the ESL includes a first ESL film having a portion contacting the sidewall surface of the dielectric layer and extending from the top surface of the dielectric layer to the top surface of the capping layer, and a second ESL film disposed over the first ESL film.

(43) In some embodiments, the semiconductor device further comprises a barrier layer surrounding the first conductive feature and disposed between the first conductive feature and the ILD layer. In some embodiments, the dielectric layer is distanced away from the first conductive feature.

(44) In some embodiments, a top surface of the ILD layer is above the top surface of the first conductive feature.

(45) Another semiconductor device comprises an interlayer dielectric (ILD) layer disposed over a substrate; a first conductive feature embedded in the ILD layer, wherein a top surface of the first conductive feature is below a top surface of the ILD layer; a dielectric layer formed on and aligned with the ILD layer; and an etch stop layer (ESL) disposed over the dielectric layer and the first conductive feature, wherein the ESL includes a first portion contacting a sidewall surface of the dielectric layer and extending from the top surface of the dielectric layer to the top surface of the first conductive feature.

(46) In some embodiments, the semiconductor device further comprises a barrier layer surrounding the first conductive feature and disposed between the first conductive feature and the ILD layer, wherein the ESL further includes a second portion disposed on a sidewall surface of the barrier layer. In some embodiments, the top surface of the first conductive feature is below a top surface of the barrier layer and further below a top surface of the dielectric layer. In some embodiments, the first conductive feature comprises a metal plug and a capping layer disposed on and aligned with the metal plug. In some embodiments, the semiconductor device further comprises a second conductive feature including a first portion landing on the first conductive feature and a second portion landing on the dielectric layer through the ESL layer.

(47) An exemplary method includes forming a first conductive feature in an interlayer dielectric (ILD) layer over a substrate; selectively depositing a capping layer on and aligned with the first

conductive feature; selectively coating a blocking layer on and aligned with the capping layer; depositing a dielectric layer over and aligned with the ILD layer, wherein the blocking layer includes a composition to prevent the dielectric layer from being deposited thereon; removing the blocking layer to expose the capping layer; and depositing an etch stop layer (ESL) on the dielectric layer and the capping layer.

(48) In some embodiments, the method further includes recessing the first conductive feature before forming the capping layer, such that a top surface of the first conductive feature is below a top surface of the ILD layer. In some embodiments, selective depositing the capping layer on and aligned with the first conductive feature includes depositing the capping layer with a thickness such that a top surface of the capping layer is below the top surface of the ILD layer.

(49) In some embodiments, selectively coating a blocking layer includes depositing a nitrogen-containing self-assembling monolayer by a chemical adsorption process. In some embodiments, depositing a dielectric layer includes depositing the dielectric layer comprising metal oxide or metal nitride by a selective atomic layer deposition (ALD). In some embodiments, removing the blocking layer includes removing the block layer by a hydrogen treatment. In some embodiments, selectively depositing a capping layer includes selectively depositing a cobalt-containing layer by a chemical vapor deposition.

(50) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device comprising: a first dielectric layer disposed on a substrate; a first conductive feature disposed in the first dielectric layer, wherein a top surface of the first conductive feature is recessed relative to the top surface of the first dielectric layer; a first barrier layer disposed in the first dielectric layer between the first conductive feature and the first dielectric layer, the first barrier layer having a sidewall surface interfacing with the first conductive feature; a second dielectric layer disposed directly on the top surface of the first dielectric layer; and an etch stop layer extending from over the second dielectric layer to the sidewall surface of the first barrier layer such that the etch stop layer interfaces with the sidewall surface of the first barrier layer.
2. The device of claim 1, further comprising: a third dielectric layer disposed on the etch stop layer; and a second conductive feature extending through the third dielectric layer, the second dielectric layer and the etch stop layer, wherein the second conductive feature is electrically coupled to the first conductive feature.
3. The device of claim 2, wherein the second conductive feature further extends within the first dielectric layer to a depth below the top surface of the first dielectric layer.
4. The device of claim 2, further comprising a second barrier layer disposed around the second conductive feature and extending through the third dielectric layer, the second dielectric layer, the etch stop layer and to the first conductive feature such that the second barrier layer interfaces with the first conductive feature and the first barrier layer.
5. The device of claim 4, wherein the second dielectric layer has a top surface facing away from the substrate, a bottom surface facing the substrate and a side surface extending from the bottom surface to the top surface of the second dielectric layer, wherein the second barrier layer interfaces with the side surface and top surface of the second dielectric layer, and wherein the first dielectric

layer interfaces with the bottom surface of the second dielectric layer.

6. The device of claim 5, wherein the etch stop layer interfaces with the top surface of the second dielectric layer.

7. The device of claim 1, further comprising a cobalt-containing layer disposed directly on the top surface of the first conductive feature such that the cobalt-containing layer interfaces with the top surface of the first conductive feature.

8. The device of claim 7, wherein the cobalt-containing layer extends to and interfaces with the sidewall surface of the first barrier layer.

9. A device comprising: a first dielectric layer disposed on a substrate; a first conductive component at least partially embedded in the first dielectric layer; a metal-containing layer disposed directly on the first conductive component; a second dielectric layer disposed on the first dielectric layer, wherein a top surface of the metal-containing layer is recessed relative to a top surface of the second dielectric layer; a first etch stop layer disposed directly on the second dielectric layer and the top surface of the metal-containing layer such that the first etch stop layer physically contacts the second dielectric layer and the top surface of the metal-containing layer; a second etch stop layer disposed directly on the first etch stop layer such that the second etch stop layer physically contacts the first etch stop layer; and a second conductive component extending through and physically contacting the second etch stop layer, the first etch stop layer, the metal-containing layer and the first conductive component.

10. The device of claim 9, wherein the second conductive component physically contacts the second dielectric layer.

11. The device of claim 9, further comprising a third etch stop layer disposed directly on the second etch stop layer such that the third etch stop layer physically contacts the second etch stop layer, and wherein the second conductive component further extends through and physically contacts the third etch stop layer.

12. The device of claim 9, wherein the first conductive component includes: a first conductive feature; and a first barrier layer disposed along an outer perimeter of the first conductive feature, and wherein the second conductive component includes: a second conductive feature; and a second barrier layer disposed along an outer perimeter of the second conductive feature.

13. The device of claim 12, wherein the metal-containing layer physically contacts the first barrier layer and the first conductive feature, and wherein the second barrier layer physically contacts the second etch stop layer, the first etch stop layer, the first barrier layer and the first conductive feature.

14. The device of claim 9, wherein the second dielectric layer physically contacts the first dielectric layer.

15. The device of claim 9, wherein the first and second etch stop layers extend within the first dielectric layer to a depth below a top surface the first dielectric layer.

16. A method comprising: providing a first conductive component disposed in a first dielectric layer over a substrate; forming a metal-containing layer on the first conductive component such that the first conductive component is covered by the metal-containing layer; forming a self-assembling monolayer on the metal-containing layer; after the forming of the self-assembling monolayer on the metal-containing layer, forming a second dielectric layer on the first dielectric layer, wherein at least a portion of a top surface of the self-assembly monolayer is exposed after the forming of the second dielectric layer on the first dielectric layer; after forming the second dielectric layer, removing at least a portion of the self-assembly monolayer to expose the metal-containing layer; and forming a second conductive component on the exposed metal-containing layer.

17. The method of claim 16, wherein the first conductive component includes: a first conductive feature; and a first barrier layer disposed along an outer perimeter of the first conductive feature.

18. The method of claim 17, further comprising recessing the first conductive feature prior to the

forming of the metal-containing layer on the first conductive component, wherein the first barrier layer extends to a greater height above the substrate than the first conductive feature after the recessing of the first conductive feature.

19. The method of claim 16, wherein the removing of at least the portion of the self-assembly monolayer to expose the metal-containing layer includes applying a hydrogen treatment process.

20. The method of claim 16, wherein the removing of at least the portion of the self-assembly monolayer to expose the metal-containing layer includes applying a nitrogen treatment process.
